

XClipGS: Exact Half-Space Clipping for Medical Volume Gaussian Splatting

Anonymous submission



Figure 1: **Faithful interactive clipping.** XClipGS, reference volume rendering, and ClipGS (Li et al. 2025a) (left to right) on a held-out test cut. The arrow marks ClipGS leakage; labels report quality and speed.

Abstract

Gaussian-splatting proxies enable interactive rendering of volumetric medical scans, but a clipping plane exposes anatomy not constrained by external-view training and intersects primitives that conventional splatting can only keep or drop whole. We present XClipGS (*eXact Clipping*), which treats these as two separate problems: the render-time clip operator and supervision of the hidden interior. Under the local affine model used by EWA splatting, each half-space-restricted Gaussian’s ray integral factorizes exactly into its ordinary 2D footprint and a conditional Gaussian CDF whose argument is affine in pixel coordinates. The resulting closed-form per-splat operator introduces no learned clipping parameters or auxiliary network and remains differentiable with respect to the primitive and plane. We use multi-distance reference views with varied clipping-plane axes and offsets to supervise the interior through the same operator. We also introduce a paired clipped/unclipped cut-face protocol with difference-referenced cut error (CDE) and culled-side leakage (Leak), because global image metrics dilute errors near the plane. On eight CT and MRI volumes with plane offsets not used for training, XClipGS attains the highest PSNR on every volume (33.56 versus 32.34 dB for ClipGS) while rendering at over 650 FPS, far above real time, versus 278 FPS. On

voxel-axis cut-face views it raises average band SSIM from .809 to .860 and leaks roughly $41\times$ less. Without retraining, it also achieves the best average across all four metrics on arbitrary-normal planes; on a fixed interior it matches RaRa’s face fidelity with about $16\times$ less leakage.

1 Introduction

CT and MRI are inspected by orbiting, zooming, and clipping the volume to expose hidden anatomy (Weiskopf, Engel, and Ertl 2003; Niedermayr et al. 2024). Cinematic volume rendering makes these cross-sections legible through realistic depth and material cues, but converged rendering remains expensive on commodity and web hardware. Gaussian splatting (Kerbl et al. 2023) offers a real-time proxy: a volume can be represented by compact primitives and rasterized at interactive rates. Existing medical proxies are usually trained from views surrounding the intact volume, however, so their supervision concentrates on the visible outer anatomy.

An interactive clipping plane changes the rendering problem. It removes the outer layers and turns a previously hidden cross-section into the dominant surface. A proxy that matches external views can therefore fail abruptly when a user moves the plane through its interior. This challenges representation generalization and requires the renderer to determine each intersected Gaussian’s partial contribution.

We separate these effects into two failure modes. The **clip**

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operator must truncate primitives intersected by the plane: binary keep/drop schemes, including ClipGS’s trainable decision (Li et al. 2025a), quantize the boundary, while RaRa’s ray–ellipsoid chord ratio (Li et al. 2025b) only approximates partial contribution. The **unsupervised interior** is equally important: an accurate operator reveals whatever anatomy the proxy learned behind the outer surface. External views can leave that anatomy ambiguous, while supervision through an approximate operator can teach the proxy the operator’s boundary errors. Faithful clipping therefore requires both an accurate partial-primitive rule and training views that reveal the hidden volume: with the exact operator in place, removing the clipped training views still costs 8.3 dB on held-out clipped views across all eight volumes.

XClipGS (*eXact Clipping*) addresses both with a closed-form, differentiable per-pixel clip and clip-aware interior supervision (Figure 1). Under the local affine model of EWA splatting (Zwicker et al. 2001), each half-space-restricted Gaussian’s ray integral equals its ordinary 2D footprint times a conditional Gaussian CDF whose argument is affine in pixel position. The clip is exact under affine EWA at the per-splat integration level and requires no learned clipping parameters or auxiliary network. Rendering the proxy through this operator against clipped reference views supplies direct gradients to primitives near the exposed face. We vary camera distance, plane axis, and offset so that training includes full-volume context and detailed, diverse cross-sections. The construction acts on primitive geometry and applies unchanged to CT and MRI within the same training and rendering pipeline.

Our experiments distinguish whole-system quality from operator quality. We compare the analytic operator with a hard cull (HC) and the forward-KL-optimal moment-matched Gaussian surrogate (MM) under a shared N-DGS backbone (Gao et al. 2025a). A 3DGS-based ClipGS (Li et al. 2025a) provides a system comparison, while operator swaps on fixed checkpoints isolate the render-time rule and include RaRa (Li et al. 2025b). Numerically held-out plane offsets test interpolation along the sampled voxel axes, while a separate test uses arbitrary plane orientations without retraining. Paired clipped and unclipped reference renders localize error at the cut face, where global metrics are least informative.

In summary, this paper contributes:

- **Analytic half-space clipping:** a closed-form, differentiable per-pixel Gaussian integral that is exact for each primitive under the affine EWA model and introduces no learned clipping parameters or auxiliary network;
- **Clip-aware interior supervision:** multi-distance reference views whose clipping-plane axes and offsets vary across the volume, with evaluation on held-out offsets and arbitrary orientations absent from training, testing transfer to new cross-sections without retraining;
- **A CT and MRI benchmark with cut-face metrics:** eight volumes with held-out planes and metrics that score the cut where global image quality is nearly blind to it, on which our method achieves the best average on all four metrics for both plane sets. Parameter sweeps verify that localization and leakage conclusions are stable across thresholds, band widths, and leakage-mask settings.

2 Related Work

Medical volume rendering and Gaussian proxies. Cinematic medical volume rendering path-traces global illumination through the volume (Kroes, Post, and Botha 2012; Comaniciu et al. 2016; Novák et al. 2018), making cross-sections legible through realistic depth and material cues (Dappa et al. 2016). Converged rendering is not interactive, progressive estimates can flicker (Martschinke et al. 2019), and web deployment remains difficult (Xu et al. 2023). Niedermayr et al. (2024) reproduce cinematic anatomy in real time with a Gaussian proxy, but its clip planes are fixed during preprocessing. Other Gaussian methods target rendering, reconstruction, or novel-view synthesis (Gao et al. 2024; Cai et al. 2024; Zha et al. 2024; Yang et al. 2024; Peng et al. 2025; Yang et al. 2026), not partial-primitive clipping.

Geometric clipping of Gaussian splats. ClipGS (Li et al. 2025a), our primary medical baseline, makes a binary per-primitive decision trainable through an STE and deforms primitives with a plane-conditioned MLP. RaRa Clipper (Li et al. 2025b) scales intersecting primitives by the visible ray–ellipsoid chord fraction, and its released formulation has no backward pass for clip-aware training. Earlier medical work toggles opacity layers (Kleinbeck et al. 2025). Our operator instead integrates the half-space-restricted Gaussian ray density under affine EWA and supports both supervision and rendering. Section 4 compares the system with ClipGS and isolates RaRa as a render-time rule on a fixed interior.

Conditioned Gaussian splatting. 6DGS (Gao et al. 2025a) and 7DGS (Gao et al. 2025b) form N-dimensional Gaussian splatting (N-DGS), with Beta-conditioned generalizations (Liu et al. 2026, 2025). Our opacity-conditioned N-DGS backbone leaves each primitive an ordinary 3D Gaussian, so geometric clipping applies unchanged; its statistical “slicing” is distinct from a planar cut. Render-FM (Gao et al. 2026) supplies initialization.

Gaussian-splatting fidelity. Our derivation adopts the local affine EWA projection used by 3DGS (Kerbl et al. 2023; Zwicker et al. 2001). Exact perspective-ray methods such as 3DGEER (Huang et al. 2026) change the projection model and require re-derived coefficients. Our exactness claim accordingly concerns the per-primitive affine-EWA ray integral; the base renderer’s splat order and alpha compositing remain unchanged. Analytic-Splatting integrates over projected pixels, while Mip-NeRF and Mip-Splatting address pixel-footprint sampling (Liang et al. 2024; Barron et al. 2021; Yu et al. 2024); these concerns are complementary to half-space restriction.

Semantic editing and half-space representations. Semantic editors select or remove whole primitive subsets (Chen et al. 2024; Ye et al. 2024; Cen et al. 2025; Zhou et al. 2024; Jain, Mirzaei, and Gilitschenski 2024). 3D-HGS instead learns a fixed half-Gaussian split (Li et al. 2025c); neither evaluates the continuous contribution of a user-supplied plane. EVSsplitting replaces both sides with moment-preserving Gaussians (Feng et al. 2024). Like our truncated-normal MM ablation (Tallis 1961), it re-Gaussianizes; our operator integrates the restricted density.

3 Method

Overview. XClipGS combines an exact per-splat render-time clipping operator with clip-aware supervision that learns the hidden interior (Figure 2). The operator analytically integrates plane-straddling Gaussians under affine EWA, while multi-distance clipped views with varied plane axes and offsets train the interior through the same differentiable rule. We first establish the rendering model, derive the analytic operator and its hard-cull and moment-matched reference points, and then describe interior supervision.

3.1 Setup and notation

A scene is a set of Gaussian primitives $\{(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, \alpha_i, \mathbf{c}_i)\}_{i=1}^N$ with mean $\boldsymbol{\mu}_i \in \mathbb{R}^3$, covariance $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3 \times 3}$, opacity $\alpha_i \in [0, 1]$, and view-dependent color $\mathbf{c}_i$ (Kerbl et al. 2023). Following the EWA formulation (Zwicker et al. 2001), a camera induces a local affine approximation of the projection at each primitive center,

$$\mathbf{u}(\mathbf{x}) \approx \hat{\mathbf{p}}_i + \mathbf{M}_i(\mathbf{x} - \boldsymbol{\mu}_i), \quad \mathbf{M}_i = \mathbf{J}_i \mathbf{W} \in \mathbb{R}^{2 \times 3}, \quad (1)$$

where $\mathbf{W}$ is the world-to-camera rotation, $\mathbf{J}_i$ the projection Jacobian at $\boldsymbol{\mu}_i$, and $\hat{\mathbf{p}}_i$ the projected center. The screen-space covariance is $\mathbf{A}_i = \mathbf{M}_i \boldsymbol{\Sigma}_i \mathbf{M}_i^\top$, and the contribution of primitive i to a pixel $\mathbf{p}$ is $\alpha_i \mathcal{K}_G(\boldsymbol{\delta}_i; \mathbf{A}_i)$ with $\boldsymbol{\delta}_i = \hat{\mathbf{p}}_i - \mathbf{p}$ and $\mathcal{K}_G(\boldsymbol{\delta}; \mathbf{A}) = \exp(-\frac{1}{2} \boldsymbol{\delta}^\top \mathbf{A}^{-1} \boldsymbol{\delta})$; contributions are alpha-composited front-to-back. This per-pixel weight is, up to normalization, the integral of the 3D Gaussian density along the pixel’s viewing ray under the affine approximation (1), an identity we exploit below. Our operator changes this per-primitive weight but retains the base rasterizer’s depth order and front-to-back compositing.

Clipping plane. A clipping plane with unit normal $\mathbf{n}$ and offset τ keeps the half-space $\mathcal{H}_\tau = \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{n}^\top \mathbf{x} \leq \tau\}$. The reference renderer removes density outside $\mathcal{H}_\tau$ before ray integration through its volumetric half-space clipping interface. XClipGS and all operator ablations accept arbitrary $(\mathbf{n}, \tau)$. For obliquely acquired scans, voxel-axis planes are generally rotated and noncanonical in world coordinates; the arbitrary-normal evaluation instead specifies planes directly in physical-world coordinates. Throughout, let

$$\sigma_n^2 = \mathbf{n}^\top \boldsymbol{\Sigma} \mathbf{n}, \quad a = \frac{\tau - \mathbf{n}^\top \boldsymbol{\mu}}{\sigma_n}, \quad \lambda(a) = \frac{\phi(a)}{\Phi(a)}, \quad (2)$$

denote the primitive’s variance along the plane normal, its standardized signed distance to the plane, and the inverse Mills ratio, where ϕ and Φ are the standard normal density and CDF. $\Phi(a)$ is precisely the fraction of the primitive’s mass on the kept side.

3.2 Half-space clipping operators

Clipping operators. A clip operator specifies how a primitive contributes when rendering under $\mathcal{H}_\tau$. The target primitive is the unnormalized restricted density $r(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) \mathbb{1}[\mathbf{x} \in \mathcal{H}_\tau]$; our operator evaluates its ray integral in closed form (Proposition 1). We contrast it with two reference points. A hard cull represents the whole-primitive

decisions used by prior work (Li et al. 2025a); a moment-matched truncation is a smooth Gaussian approximation derived from classical truncated-normal statistics. Comparing them in one renderer isolates the cost of binary truncation and re-Gaussianization relative to integrating the restricted density itself. We abbreviate the hard cull, moment-matched truncation, and our analytic operator as HC, MM, and Ours.

Hard cull (HC). This operator keeps or drops each primitive *whole* using $\mathbb{1}[\mathbf{n}^\top \boldsymbol{\mu} \leq \tau]$. It adds no per-pixel factor, but straddling primitives overshoot or vanish, producing a center-quantized boundary as the plane sweeps. The raw decision is also non-differentiable. ClipGS trains its per-primitive decision through a straight-through estimator and deforms primitives; HC isolates the underlying binary operator from those learned components.

Moment-matched truncation (MM). This surrogate, which we derive from classical statistics rather than take from prior work, replaces each straddling primitive by the Gaussian whose first two moments match the normalized truncated density $r/\Phi(a)$. Equivalently, it minimizes $D_{\text{KL}}(r/\Phi(a) \parallel q)$ over Gaussian q . The required one-sided truncated-normal moments (Tallis 1961), propagated through the joint covariance, use $\mathbf{d}_n = \boldsymbol{\Sigma} \mathbf{n} / \sigma_n$ and are

$$\begin{aligned} \alpha' &= \Phi(a) \alpha, & \boldsymbol{\mu}' &= \boldsymbol{\mu} - \lambda(a) \mathbf{d}_n, \\ \boldsymbol{\Sigma}' &= \boldsymbol{\Sigma} + [v(a) - 1] \mathbf{d}_n \mathbf{d}_n^\top. \end{aligned} \quad (3)$$

where $v(a) = 1 - a \lambda(a) - \lambda(a)^2$. Opacity scales by the kept mass, the mean shifts to the truncated centroid, and the variance contracts along the clip normal by the truncated-variance factor $v(a) \in (0, 1]$. MM is smooth in $(\boldsymbol{\mu}, \boldsymbol{\Sigma}, \alpha, \tau)$ and can supervise the cut face directly. Its limitation is intrinsic: re-Gaussianization restores unbounded support, so some mass still renders beyond the plane and the cut becomes a smooth tail of width $\mathcal{O}(\sigma_n)$ rather than an edge.

Exact per-pixel integral (Ours). We render the truncated density itself rather than re-Gaussianizing it. Under the affine approximation (1), screen position $\mathbf{u}$ and clip coordinate $\omega = \mathbf{n}^\top \mathbf{x}$ are jointly Gaussian. Conditioning on $\mathbf{u} = \mathbf{p}$ factorizes the truncated ray integral into the unclipped footprint and $P(\omega \leq \tau \mid \mathbf{u} = \mathbf{p})$. With $\mathbf{b} = \mathbf{M} \boldsymbol{\Sigma} \mathbf{n} \in \mathbb{R}^2$, the conditional variance is

$$s^2 = \sigma_n^2 - \mathbf{b}^\top \mathbf{A}^{-1} \mathbf{b}, \quad (4)$$

and the clipped per-pixel contribution becomes

$$\boxed{\begin{aligned} &\alpha \mathcal{K}_G(\boldsymbol{\delta}; \mathbf{A}) \cdot \Phi(\ell(\boldsymbol{\delta})), & \ell(\boldsymbol{\delta}) &= \mathbf{k}^\top \boldsymbol{\delta} + h, \\ &\mathbf{k} = \frac{\mathbf{A}^{-1} \mathbf{b}}{s}, & h &= \frac{\tau - \mathbf{n}^\top \boldsymbol{\mu}}{s}. \end{aligned}} \quad (5)$$

Up to normalization, this Gaussian–CDF product has the classical selection-normal form (Arellano-Valle, Branco, and Genton 2006). We specialize it to half-space EWA ray integration and derive the rasterizer coefficients and gradients. The clip thus reduces to three transient coefficients $(\mathbf{k}, h)$ computed from existing geometry per primitive and view, plus one CDF evaluation per sample with argument affine in the pixel offset $\boldsymbol{\delta}$ already available to the rasterizer. These are render-time intermediates, not learned parameters.

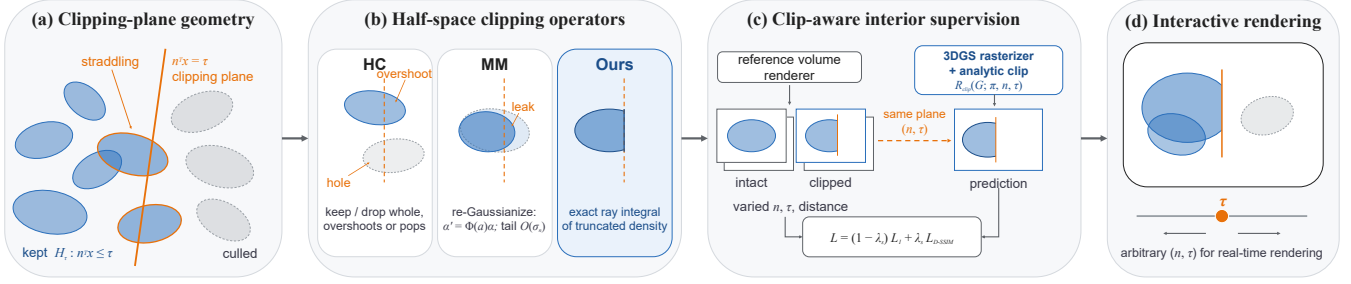


Figure 2: **XClipGS pipeline.** Exact per-splat half-space integration clips straddling Gaussians, while multi-distance cut views supervise the hidden interior through the same differentiable operator.

Why the factor is per pixel. The marginal kept-mass fraction $\Phi(a)$ from Equation 2 is a single scalar for the whole primitive. Multiplying the projected footprint uniformly by this value would attenuate the splat but could not locate the cut within it. Equation 5 instead uses the conditional probability given the pixel. The term $\mathbf{k}^\top \delta$ moves the transition across the footprint, while h shifts it as the plane moves. Consequently, pixels whose rays pass through the retained side approach a factor of one and those on the removed side approach zero, with the transition width determined by the remaining uncertainty s along the ray.

Proposition 1 (Exactness). *Under the local affine approximation (1), the contribution (5) equals the viewing-ray integral of the unnormalized half-space-restricted density r , up to the global normalization shared with the unclipped splat.*

Conditioning the jointly Gaussian screen and clip coordinates on $\mathbf{u}=\mathbf{p}$ factorizes the ray integral into the EWA footprint and $P(\omega \leq \tau \mid \mathbf{u}=\mathbf{p})$. Its Schur-complement variance is Equation 4, and its affine conditional mean yields Equation 5. Supplement S1 gives the full derivation.

Scope and boundary behavior. Proposition 1 is exact for one primitive’s ray integral within the affine EWA model (1) (Zwicker et al. 2001) and adds no approximation to its unclipped footprint. At image level, XClipGS retains the base rasterizer’s standard center-based splat order and front-to-back alpha compositing; the proposition isolates the added half-space integral from these established renderer choices. A perspective-ray formulation requires re-derived conditional coefficients (Huang et al. 2026). Within affine EWA, the boundary width is $s \leq \sigma_n$; as the plane becomes parallel to the rays, $s \rightarrow 0$ and $\Phi(\ell)$ approaches a per-pixel step without MM’s tail.

Differentiability. For $s > 0$ our operator is smooth in primitive parameters and in τ ; at the degenerate $s \rightarrow 0$ limit the factor becomes a step (a plane parallel to the ray). Since $\partial_\ell \Phi = \phi(\ell)$, gradients pass through $(\mathbf{k}, h)$ and the center to (μ, Σ) . Thus MM and Ours supervise the cut itself. HC has no gradient through its binary decision and updates only kept primitives, the discontinuity ClipGS addresses with an STE.

3.3 Clip-aware interior supervision

Unconstrained interiors. A proxy trained only from external views leaves hidden density weakly constrained: primitives behind the visible surface receive gradients only through

accumulated transmittance, and the anatomy exposed by a new cut was never observed directly. Interior fidelity therefore depends on supervision as well as the clip operator. The same external images can be explained by different interior configurations that disagree once the front layers are removed. Clipped views resolve this ambiguity by directly exposing hidden density, not by adding cameras around the intact outer surface alone. The analytic operator makes the cut itself differentiable, turning interior supervision into ordinary photometric training: gradients from clipped reference views flow through Φ to primitives at the exposed face.

Training-view generation. For each scan, the reference volume renderer generates both intact and clipped views across a range of camera distances. Intact views orbit and frame the whole scan. For clipped views, we sample one of the three voxel axes and an offset $\tau \sim \mathcal{U}[\tau_{\min}, \tau_{\max}]$ spanning the volume core, then aim the camera obliquely at the exposed face. Each frame records its plane $(\mathbf{n}, \tau)$, and the same plane is passed to the rasterizer during training. Varying distance supplies both full-volume context and close-up detail; varying axis and offset exposes diverse cross-sections rather than a single prescribed cut.

Training objective. Training minimizes the standard 3DGS photometric loss through the differentiable clip (Kerbl et al. 2023; Wang et al. 2004),

$$\mathcal{L} = (1 - \lambda_s) \mathcal{L}_1(\mathcal{R}_{\text{clip}}(\mathcal{G}; \pi_v, \mathbf{n}_v, \tau_v), I_v) + \lambda_s \mathcal{L}_{\text{D-SSIM}}, \quad (6)$$

where $\mathcal{R}_{\text{clip}}$ renders the primitive set $\mathcal{G}$ under camera π_v at plane $(\mathbf{n}_v, \tau_v)$ ($\tau_v = \infty$ for intact views), I_v is the reference render, and $\lambda_s = 0.2$ in all experiments. Gradients from the exposed cross-section flow through Φ into the position, covariance, and opacity of interior primitives near the cut. Section 4 tests both held-out offsets along the training axes and arbitrary plane orientations without retraining.

3.4 Implementation

Rasterization. Our CUDA implementation combines Speedy-Splat culling (Hanson et al. 2025), TC-GS (Liao et al. 2025), and Mip-Splatting’s 3D filter (Yu et al. 2024). Preprocessing computes the shared truncation statistics. MM then rewrites each straddling primitive once, whereas Ours retains its ordinary EWA footprint and evaluates $\Phi(\ell)$ only for samples from straddling primitives. Both paths remain

Table 1: **Held-out quality and speed on eight volumes.** FPS is measured at 1600×1600 on an A100. **Bold** marks the best.

Scene	XClipGS (Ours)				ClipGS (reimpl.) (2025a)				MM				HC			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FPS $\uparrow$
abdomen	33.30	.965	.083	682	32.20	.960	.089	293	32.97	.964	.084	615	33.02	.963	.085	651
intestine	31.77	.949	.093	597	30.83	.944	.100	220	31.49	.948	.094	575	31.50	.947	.095	599
knee-joint	29.14	.913	.185	586	27.89	.904	.197	252	28.85	.912	.186	580	28.44	.908	.191	604
lower-limb	34.28	.980	.026	670	33.02	.977	.028	241	34.20	.980	.026	653	34.04	.980	.026	693
vascular	36.38	.975	.031	666	34.68	.971	.035	320	36.18	.975	.031	621	36.10	.974	.032	654
heart	29.20	.940	.095	603	28.14	.930	.106	239	28.72	.938	.097	579	28.65	.936	.101	594
nose (MRI)	35.92	.977	.047	699	34.65	.973	.051	306	35.67	.976	.048	698	35.55	.975	.049	694
hand (MRI)	38.48	.978	.036	740	37.30	.974	.041	353	38.22	.977	.037	724	37.85	.975	.040	772
avg	33.56	.960	.075	655	32.34	.954	.081	278	33.29	.959	.075	631	33.14	.957	.077	658

differentiable end to end and accept arbitrary plane orientations. Supplements S2–S3 give numerical safeguards, gradient validation, and experimental configurations.

4 Experiments

We evaluate held-out plane offsets and orientations, cut-face errors, and render-time operator swaps. Ours, MM, and HC share the N-DGS backbone, initialization, views, and planes; ClipGS uses the same initialization, views, planes, and iteration budget with its 3DGS representation.

4.1 Experimental protocol

Datasets and splits. The benchmark comprises eight clinical volumes: six CT (abdomen, intestine, knee-joint, lower-limb, heart, vascular) and two MRI (nose, hand), spanning smooth and high-frequency anatomy. A physically based Monte Carlo volume path tracer produces a **general** set of 900 views (810 train/90 test). Camera distance varies; half of the views are intact and half contain a plane sampled across the volume core. Test-plane offsets are numerically disjoint from training and drawn from the same interval, evaluating interpolation to held-out cuts throughout the sampled volume core. A disjoint voxel-axis **cut-eval** set uses three fixed planes per volume and 30 cameras, each rendered both with and without clipping, yielding the paired references used by the localized metrics. A second set evaluates arbitrary-normal center planes without retraining. Supplement S3 specifies volume preparation, training-view sampling, and rendering; Supplement S5 specifies both cut-eval plane sets. The general set mixes fit-to-frame views with close-ups: intact views supply the outer anatomy and whole-volume context, while clipped cameras target the exposed face from oblique angles. The train/test split is stratified by view mode and plane axis, so each regime is represented without reusing an exact offset.

Comparison methods. **Ours** is the analytic half-space clip of Section 3.2. **MM** (forward-KL-optimal Gaussian surrogate) and **HC** (bare per-primitive hard cull) share Ours’ N-DGS backbone, initialization, planes, and supervision, isolating the operator within one representation. HC corresponds to the binary operator class without ClipGS’s STE training or deformation network. **ClipGS** (Li et al. 2025a) is our best-effort system-level reimplementation from its published description, using its vanilla-3DGS back-

bone, trainable visibility, and plane-conditioned deformation MLP; no public code was available. **RaRa** (Li et al. 2025b) enters only as a render-time operator on fixed interiors (Section 4.4). Supplement S4 details both baselines. Ours, MM, HC, and ClipGS are trained for 30,000 iterations, and every result uses the fixed final checkpoint.

Evaluation metrics. On general views we report PSNR, SSIM (Wang et al. 2004), LPIPS (Zhang et al. 2018), and FPS. Cut-eval adds four cut-face metrics. S_b is band SSIM over the exposed cut face. The *difference-referenced cut error* CDE measures *where* the cut acts: for the method and reference, we subtract the clipped render from its unclipped companion, threshold the resulting removal map, and normalize the symmetric-difference area by the reference region. This construction cancels reconstruction error common to both renders and penalizes both missed cuts and over-removal. We report it over the exposed face (CDE_p) and grazing edge (CDE_g). Leak is the fraction of near-plane foreground energy on the culled side, where the reference is empty. As an image-space reference metric, it measures agreement with the independent path-traced reference and also captures proxy and renderer mismatch. Supplement S5 gives the equations, per-volume results, parameter sensitivity, and the camera-independent wrong-side-mass diagnostic $CErr_{3D}$, which measures operator geometry and is zero for Ours by analytic definition. Band PSNR differs by only hundredths and is omitted.

4.2 Held-out quality and speed

Image quality. Table 1 reports held-out quality and speed. Our method has the highest PSNR on every volume and improves over the ClipGS reimplementation by +1.22 dB on average PSNR. This is a system-level comparison because ClipGS uses the vanilla 3DGS representation specified in its paper. The controlled N-DGS ablations trace the operator effect: the smooth surrogate (MM, 33.29 dB) and the bare hard cull (HC, 33.14) both trail Ours (33.56), with the same overall trend in SSIM and LPIPS. Because every evaluation offset is disjoint from training, these scores show that the trained system maintains fidelity when interpolating to numerically held-out cross-sections along the sampled axes. The advantage also holds across all eight volumes rather than being driven by one scan: Ours leads PSNR in every row, from smooth soft tissue to thin vessels and fine joint structure.

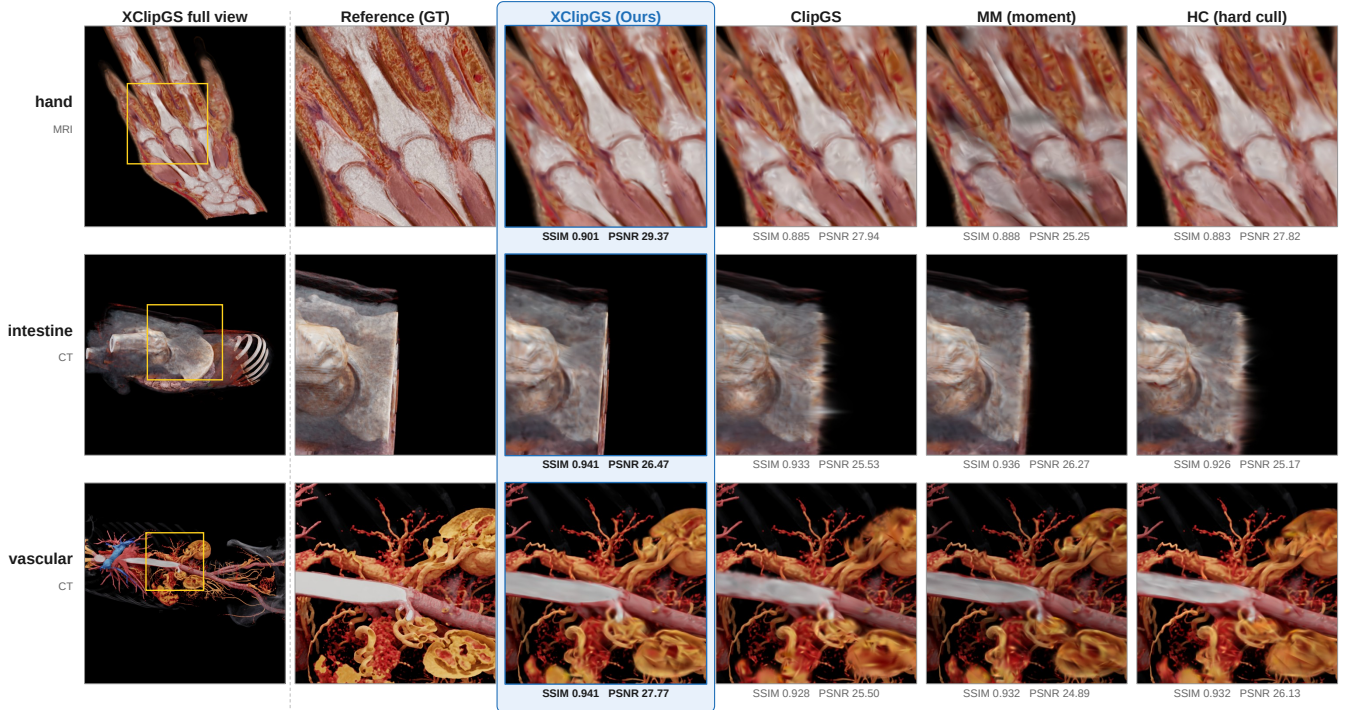


Figure 3: **Held-out qualitative comparison.** Full views locate each crop; the middle row is an unseen arbitrary-normal grazing view. The remaining columns compare matched reference and method crops, with SSIM and PSNR shown below.

Global metrics nevertheless dilute the operator effect, because half of the held-out views contain no cut and, in the remaining views, the boundary occupies only a small image region. The largest Ours–HC gap occurs on knee-joint (+0.70 dB), which contains fine cut-face structure. This dilution motivates the localized evaluation in Section 4.3.

Rendering speed. The analytic cut adds little measured overhead: Ours renders at 655 FPS, matching the hard cull (HC, 658) that does no per-sample work and the moment-matched surrogate (MM, 631), with all three within a few percent. Thus, the closed-form half-space factor adds negligible cost at this resolution on the measured A100. The ClipGS reimplementation runs at 278 FPS ($\sim 2.4\times$ slower), reflecting its extra plane-conditioned network per frame. The controlled Ours/MM/HC comparison shows the analytic operator itself runs at approximately hard-cull speed rather than trading fidelity for interactivity.

4.3 Cut-face fidelity and localization

Cut-eval protocol. To measure the clip where the global metric dilutes it, the cut-face metrics act on the held-out cut-eval set: three fixed planes per volume (one per voxel axis) at the volume core, each viewed by five *perpendicular* cameras (looking down the normal, so the cut face fills the frame) and five *grazing* cameras (tilted 1.5° off edge-on, exposing the cut edge). The clipped ground truth is rendered by the reference volume renderer with the plane applied by the renderer itself, and each camera also renders an *unclipped* companion for the difference-referenced error, supplying a matched volumetric reference for the partial-clipping behavior prior

work calls “hard to define” (Li et al. 2025b) (details in Supplement S5). To test orientation generalization, a second set samples five arbitrary physical-world normals per volume, each at least 20° from every voxel axis and from the other sampled normals. Each plane passes through the volume center and is viewed once perpendicularly and twice at grazing incidence, giving 15 views per volume. We evaluate the same checkpoints without retraining.

Image-space results. Table 2 and Figure 3 expose differences diluted by global metrics. XClipGS has the best average on all four metrics for both plane sets and the highest voxel-axis band SSIM on seven volumes. Without retraining, its arbitrary-normal averages are $.874 S_b$, $23.5 CDE_g$, $14.3 CDE_p$, and 0.13×10^{-2} Leak. It also has the highest displayed SSIM and PSNR across the examples, which span fine MRI structure in hand, layered soft tissue in intestine, and thin vessels in vascular. XClipGS preserves their structure and contrast while avoiding the diffuse or jagged boundaries of the approximations. Supplement S5 gives the per-volume results and arbitrary-normal $CERR_{3D}$.

Because these orientations and camera-plane configurations never appear in training, this set tests whether axis-supervised interiors support new center-plane orientations. The preserved ordering also shows that the analytic operator is not tied to voxel axes.

On voxel-axis cuts, Ours reduces CDE_g to 22.7, versus 23.6 for MM, 29.4 for HC, and 33.4 for ClipGS; Leak falls to 0.40×10^{-2} , versus 6.18, 12.54, and 16.42. MM retains an unbounded tail, while HC keeps or removes whole straddling splats. Supplement S5 confirms the lowest face CDE at all

Table 2: **Average cut-face results over eight volumes.** S_b : band SSIM; CDE_g/CDE_p : edge/face error; Leak: leakage. Errors are $\times 10^{-2}$; **bold**: best per row.

Plane set	XClipGS (Ours)				ClipGS (reimpl.) (2025a)				MM				HC			
	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$
Voxel-axis	.860	22.7	17.9	0.40	.809	33.4	19.5	16.42	.853	23.6	18.4	6.18	.819	29.4	18.5	12.54
Arbitrary-normal	.874	23.5	14.3	0.13	.844	35.9	15.7	17.24	.870	25.5	15.1	5.67	.842	35.9	14.9	18.30

Table 3: **Fixed-interior operator swap.** Each row changes only the clip rule on the voxel-axis set; metrics follow Table 2. **Bold**: full-precision row best; per-volume: Supplement S7.

Fixed interior	Ours (analytic)				RaRa (2025b)				MM				HC			
	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$	$S_b \uparrow$	$CDE_g \downarrow$	$CDE_p \downarrow$	Leak $\downarrow$
Ours-trained	.860	22.7	17.9	0.40	.856	23.4	18.3	6.24	.848	23.7	18.0	5.85	.838	30.1	18.2	14.89
HC-trained	.828	23.1	18.5	0.37	.826	23.6	18.8	4.89	.828	23.5	18.4	5.12	.819	29.4	18.5	12.54

four thresholds and Leak in all 27 settings; Ours wins edge CDE in 11 of 16 settings and MM in five.

Global versus geometric fidelity. MM nearly matches Ours in global PSNR and SSIM, yet its unbounded tail produces about $15.5\times$ more leakage; HC is inexpensive but raises edge error. Our factor preserves the ordinary footprint and changes only the half-space mass, explaining its fidelity, speed, and cleaner boundary. Exactness is per-splat and operator-level: Equation 5 integrates each represented Gaussian inside the half-space, while projection, ordering, and compositing remain inherited from the base renderer. $CERR_{3D}$ is zero by construction even for arbitrary normals. Image-space Leak instead compares the learned proxy with an independent path-traced reference, so its small residual reflects density, color, and compositing mismatch rather than outside-plane mass from our operator.

4.4 Fixed-interior operator comparison

Operator-swap protocol. We render fixed checkpoints through each rule on the voxel-axis set, decoupling operator and interior. This admits RaRa (Li et al. 2025b), whose released rule lacks a backward path; its renderer agrees with ours to 42.8 dB on unclipped views, indicating small off-plane differences. The jointly trained ClipGS cull and deformation network remain in the system-level comparison above, where the complete learned system is evaluated.

Operator-swap results. On the Ours-trained interior, our operator has the best average CDE_g/CDE_p (22.7/17.9) and lowest Leak (0.40, versus 6.24 for RaRa, 5.85 for MM, and 14.89×10^{-2} for HC; Table 3). RaRa nearly matches band quality but leaks about $16\times$ more. This agrees with the analytic distinction: a chord fraction is not the Gaussian mass retained along the ray, whereas Equation 5 evaluates that mass in closed form at every covered pixel.

Table 2 additionally evaluates MM and HC on their co-adapted interiors, while the fixed-interior test supplies RaRa with the same clip-aware interior as the other rules. Repeating the swap on the HC-trained checkpoint preserves Ours’ average CDE_g and Leak lead across both interiors. Supplement S7 gives per-volume results; Supplement S4 details renderer agreement.

4.5 Clip-aware supervision ablation

We fix the clip rule and vary only supervision. The *external-view-only* control shares the backbone, initialization, test split, and 30,000-iteration budget but trains only on intact views. It gains 0.7 dB on intact views yet loses 8.3 dB on every volume’s clipped views ($32.0 \rightarrow 23.8$). Voxel-axis band SSIM falls from .860 to .346 and CDE_p rises from 17.9 to 25.6×10^{-2} . The 8.3 dB gain from clip-aware supervision therefore quantifies its contribution to interior fidelity.

Together, the two controls isolate complementary effects: the operator swap attributes lower leakage to analytic clipping, while the supervision ablation holds that operator fixed and attributes interior fidelity to clipped views. Faithful clipping needs both (Supplement S6, Table S5).

4.6 Discussion

XClipGS separates responsibilities: clip-aware supervision learns the exposed anatomy, while the analytic renderer enforces each requested half-space. At deployment, a plane changes only transient per-primitive coefficients and one conditional-CDF factor per covered sample; one checkpoint therefore supports moving voxel-axis or oblique cuts without a plane-conditioned network or retraining. This separation also keeps the viewer interface simple: intact and clipped rendering share one learned volume and appearance model while the user updates only the requested plane.

5 Conclusion

XClipGS couples an exact closed-form per-splat affine-EWA integral with photometric interior supervision through the same differentiable operator, without learned clipping parameters or an auxiliary network. Across eight CT and MRI volumes, it leads all four cut-face metrics on voxel-axis and arbitrary-normal planes at hard-cull speed. The 8.3 dB advantage from clipped training views confirms that faithful clipping benefits from both analytic truncation and a supervised interior. Paired clipped/unclipped evaluation localizes boundary error, while arbitrary-normal tests demonstrate transfer beyond the supervised axes. Because new half-spaces require no retraining, XClipGS provides a reusable primitive for interactive Gaussian volume viewers.

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